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A robust hybrid approach with product context-aware learning and explainable AI for sentiment analysis in Amazon user reviews

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Abstract
In the ever-changing world of business, gaining valuable insights from customer perspectives is crucial. Consumer evaluations are crucial performance indicators for businesses seeking to enhance their impact. Cyberspace is expanding with an increasing volume of reviews, making it challenging to extract relevant information for desired products. This research explores sentiment analysis for Amazon product reviews in the domain of communication technology, utilizing four publicly available datasets. Sentiment analysis is frequently employed to support E-Commerce platforms in monitoring customer feedback on their products and striving to understand customer needs and preferences. Acknowledging that solely relying on user reviews is insufficient to achieve the best performance, we enhance our approach by incorporating additional context from product titles and headlines for a more comprehensive understanding of the learning algorithm. This paper utilizes three distinct embedding methods, including TF-IDF, Word2Vec, and FastText. FastText outperformed other embeddings when stacked with XGBoost and CatBoost, resulting in the FastXCatStack model. This model achieved accuracy scores of 0.93, 0.93, and 0.94 on mobile electronics, major appliances, and personal care appliances datasets respectively, and linear SVM showed an accuracy score of 0.91 on software reviews when combined with FastText. This research study also provides a comprehensive analysis of deep learning-based models, including approaches like LSTM, GRU, and convolutional neural networks as well as transformer-based models such as BERT, RoBERTa, and XLNET. In the concluding phase, interpretability modeling was applied using Local Interpretable Model-Agnostic Explanations and Latent Dirichlet Allocation to gain deeper insights into the model's decision-making process.

Keywords Machine learning · Deep learning · Product context · Interpretability modeling · Transformers · Word embedding

Extended author information available on the last page of the article

1 Introduction

In the era of rapid advancements in information and communication technologies, there is no disputing that social media has evolved into an essential aspect of people's daily lives in modern society [1]. The expansion of user-generated content has been generated by the rise of the internet, allowing users to express themselves and participate in concerns across multiple platforms such as blogs, social networks, e-commerce sites, and forums and this democratization of content creation has enabled individuals to express their views on a wide range of topics [2]. Additionally, the digital transition has not only impacted communication but also traditional media surroundings, enabling a more diverse and dynamic rush of ideas in cyberspace [3]. It is important for individuals, organizations, and governments to extract and utilize essential information from this data [4, 5]. With the increasing volume of data, the task of gathering pertinent information promptly and efficiently becomes more significant. This highlights the value of employing computational linguistic approaches. While previous studies have extensively explored Sentiment Analysis (SA) within the realm of e-commerce, this paper seeks to bridge the gap by focusing specifically on the integration of product context-aware learning and Explainable AI (XAI) for a more nuanced SA. By leveraging user-generated content, this research addresses the critical need for more precise and interpretable sentiment assessments in e-commerce environments. This approach not only enhances the accuracy of SA but also offers deeper insights into consumer behavior, which is essential for businesses looking to adapt to consumer needs dynamically. SA is an automated method that extracts users' sentiments, feelings, and opinions from raw human text [6]. It is a subject of study within the broader domain of Natural Language Processing (NLP) [7]. One method for SA is to use text categorization, in which texts are organized according to their sentiment perspective using a Machine Learning (ML) methodology [8]. SA has found application in diverse fields, such as healthcare [9] and product reviews [10], and more key areas.

Recent advancements in Artificial Intelligence (AI) and NLP have increased the popularity of SA, leading to the development of numerous innovative methods for research in this field [11]. Individuals utilize diverse social platforms to share their opinions regarding various entities, which can involve products or any other subject. There are some top well-known US brands, recognized for their high public demand and reputation, such as eBay, Walmart, Amazon, and Apple. This study focuses particularly on Amazon product reviews, where customers express their feedback through star ratings, providing recommendations for potential end-users and important knowledge for organizations seeking improvement [12]. Amazon[1] stands as one of the globe's foremost e-commerce platforms and ranks among the largest IT companies globally. Amazon operates online, enabling customers to place orders effortlessly through its app or website. Amazon offers a wide range of products, including health supplements, gaming items, and electronics, showcasing its

[1] https://www.amazon.com/ref=nav_logo.

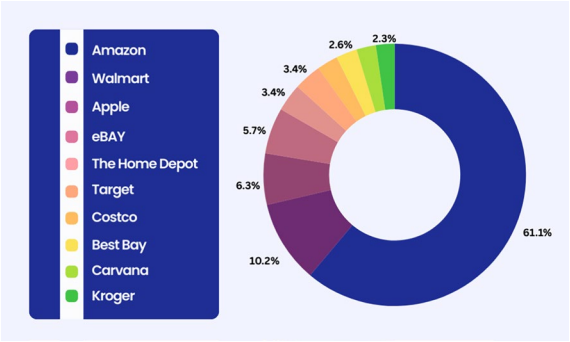

Fig. 1 US retailers' 2022 eCommerce market share

extensive potential [13]. Moreover, this business model isn't exclusive to Amazon; independent sellers also have the opportunity to emulate a similar approach. The following Fig. 1 illustrates the market share of e-commerce in the United States' top retailers for the year 2022.[2]

This paper seeks to extract sentiments from Amazon user reviews by incorporating product context-aware learning through ML-based methods, Deep Learning (DL), and transformer-based models with hyperparameter tuning. Following, the contributions of this research work are summarized, followed by how the rest of the paper is organized.

This study makes several important contributions to the field of SA. We utilize four publicly available Amazon product review datasets from distinct categories, enhancing them with product titles and headlines to improve the contextual foundation for SA. Our research systematically explores and compares the effectiveness of various word embeddings, including Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec, and FastText, with FastText showing superior performance. By applying these embeddings to a variety of ML-based models, including Support Vector Machine (SVM), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Categorial Bossting (CatBoost), we develop our innovative FastXCatStack model. Further, we enhance our approach by integrating FastText with DL-based models like Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Convolutional Neural Networks (CNNs), and incorporate advanced transformer-based models such as Bidirectional Encoder Representation for Transformers (BERT), Robustly Optimized BERT (RoBERTa), and Auto-Regressive Transformer (XLNET), each optimized through meticulous hyperparameter tuning. Based on the best performance of FastXCatStack across all datasets, we employed the XAI with Local Interpretable Model-Agnostic Explanations (LIME) and topic modeling with Latent Dirichlet Allocation (LDA). Experimental evaluations were further validated through rigorous experimental evaluations, assessing metrics such as coherence, perplexity, accuracy, F1-score, precision, and recall.

[2] https://capitaloneshopping.com/research/amazon-statistics/.

The rest of the paper is structured as follows: Sect. 2 discusses the existing research on SA for online product reviews. Section 3 explains the proposed methodology. Section 4 focuses on the results and discussions. Section 5 explains the comparison of our work with the baseline studies. Section 6 is related to the interpretability modeling. Section 7 presents the conclusion and future work.

2 Related work

In this section, we will discuss the latest advancements in the field of SA for Amazon product reviews, extensive research has explored methodologies ranging from traditional ML to advanced DL-based models.

2.1 ML based methods in SA

Hossain et al. [10] conducted tertiary classification on customer reviews of insurance products using supervised ML approaches. The dataset was collected from Yelp[3] resulting in an accuracy score of 0.94 when employing the LR classifier. Punetha et al. [14] proposed an innovative Bayesian mathematical framework for SA of reviews grounded in Game Theory. Their study focused on addressing the binary classification problem across three distinct datasets: food, mobile, and electronics. Their research study comprehended each review and extracted context scores from the review comments using the SentiWordNet lexicon [15]. By employing Part-of-Speech (POS) tagging [16], they calculated the context score, and subsequently, both the context score and rating score were input into their novel Bayesian model to predict the sentiment tag and sentiment of each review.

Abbas et al. [17] developed a supervised ML method to spot defective products in online reviews, using a dataset of Amazon reviews across four product categories. They applied statistical techniques like correlation [18] and information gain [19] for feature selection, focusing on features like emotional tone and emotions. Their best results came from the Random Forest classifier, with a 0.84 accuracy score. Meanwhile, Ahmed et al. [20] proposed a hybrid ML framework to analyze customer-service provider interactions, aiming to predict sentiment changes. They examined 5000 conversations from Twitter, targeting English tweets about "@ AmazonHelp".

Al-Obeidi et al. [21] introduced a novel 7D hyperchaotic system, positing significant advancements in synchronization techniques. Subsequent scrutiny led to the retraction of this paper due to identified irregularities, notably in peer review and data integrity. This development highlights the complexities and challenges inherent in maintaining robustness and reliability in computational neuroscience research. Similarly, Alsaffar et al. [22] worked on a network management system for IoT using dynamic systems, which provided insights into IoT device management and

[3] https://www.yelp.com/dataset/download.

network performance enhancements. In recent advancements in the mathematical modeling of complex systems, the work conducted by Hameed et al. [23] expands on the foundational concepts of limit superior and limit inferior in the context of soft sequences. This contributes to our analysis by providing a nuanced approach to interpreting the upper and lower bounds in datasets characterized by inherent ambiguities, similar to those we encounter in SA of consumer reviews.

2.2 DL and transformers based approaches for SA

Alantari et al. [24] conducted an empirical analysis of SA using supervised ML, DL, and transformer-based approaches. They explored 25,241 products across nine categories and analyzed 260,489 reviews from five different sites. According to their findings, pre-trained neural network-based methods yielded the most accurate predictions. Additionally, their study highlighted that topic modeling, specifically LDA [25], offered informative details. In their study, Huang et al. [26] implemented SA of Chinese e-commerce product reviews using Enhanced Representation through Knowledge Integration (ERNIE) [27] word embedding with Bidirectional LSTM (BiLSTM). The embeddings extracted from ERINE were fed into a BiLSTM, followed by the application of an attention mechanism to standardize and optimize the weights. The final prediction was obtained by passing these optimized weights through a softmax layer and providing them with the F1-score of 0.87. Suhartono et al. [28] introduced a tertiary classification problem for the SA of drug product reviews, employing Deep Neural Networks (DNNs) and weighted word embeddings. Two distinct word embedding techniques, GLOVE [29] and Word2Vec [30] were utilized, with the embeddings passed through various layers of a Convolutional Neural Network (CNN) [31]. Additionally, transformer-based methods, including BERT [32] and RoBERTa [33], were implemented. Among these, their CNN model utilizing GLOVE word embeddings achieved the highest accuracy, reaching 0.85.

A similar approach was addressed by Qorich et al. [34] for the text sentiment classification on Amazon reviews through the integration of word embeddings and CNN. They employed two distinct word embedding methods, FastText [35] and Word2Vec, and applied their approach to three diverse datasets. Notably, their proposed CNN-based model exhibited superior performance compared to the baseline across all datasets, surpassing both traditional ML and DL-based methods. The CNN model demonstrated enhanced accuracy, reaching 0.90 specifically on the Amazon dataset, highlighting its effectiveness in SA tasks. Bilal et al. [36] introduced a supervised approach and fine-tuned a BERT model for the classification of helpful and unhelpful online customer reviews. Their study utilized the Yelp shopping reviews dataset for evaluation. Notably, the fine-tuned BERT base model achieved the highest accuracy score, reaching 0.70.

In recent developments within digital security techniques, significant advances have been made to enhance the capacity and undetectability of text-based steganography. Azzam et al. [37] an innovative approach by using binary digit mapping on ASCII characters to create high-capacity, undetectable text steganography, highlighting potential applications in securing e-commerce

communications. This methodology highlights the evolving complexity of digital security measures, relevant to our discussion on the enhancement of data privacy and security in online platforms. In exploring innovative materials for enhancing the physical infrastructure of e-commerce logistics, recent research offers valuable insights. Abdulla et al. [38] investigated the toughness of timber beams reinforced with jute fibers, demonstrating significant improvements in structural resilience. Their findings indicate that jute fiber-reinforced timber could offer enhanced durability and load-bearing capacity, which are critical factors in the development of robust e-commerce warehousing solutions. In light of the evolving requirements for durable and resilient infrastructure in e-commerce logistics and warehousing, the study by Abdullah et al. [39] offers critical insights into the use of advanced material technologies. Their research, which examines the flexural behavior of box ferrocement beams reinforced with fiberglass mesh and GFRP bars post-exposure to high temperatures, highlights the potential of these materials to enhance structural integrity under adverse conditions. This aligns with our discussion on leveraging innovative materials to improve the reliability and efficiency of logistics systems in digital commerce environments, demonstrating that such advancements not only support structural durability but also contribute to the sustainability of e-commerce operations.

Enríquez et al. [40] proposed a transformer-based SA for Mexican tourist reviews, focusing on key locations. The dataset included reviews in Spanish and English, with a multi-class classification task (0–5) on hotels, restaurants, and Attractions. Addressing data imbalance, they used SVM with FastText and explored transformer models like RoBERTaESP and GPT-2 [41]. Naseem et al. [42] introduced a transformer-based methodology for binary classification of English tweets, utilizing the Transformer Deep Intelligent Contextual Embedding (DICET) framework. This framework comprises three primary components: an intelligent preprocessor, a text representation layer, and a Bi-directional Long- Short-Term Memory (BiLSTM), along with various word embeddings such as GLOVE, POS, and lexicon-based embeddings, integrating with attention modeling. Their study involved the utilization of three distinct datasets.

In the study conducted by Hashir et al. [43], a comprehensive SA of Amazon product reviews was performed using various ML, ensemble learning, and DL-based methods. The study meticulously employed NLP techniques, from data preprocessing steps like tokenization and vectorization through advanced analytical methods employing transformers such as BERT and XLNet. Their evaluations demonstrated that the BERT model achieved an exceptional accuracy of 89%, highlighting its effectiveness in deriving actionable insights from vast consumer feedback data. This work not only underscores the evolving complexity in sentiment analysis tasks but also sets a benchmark in integrating DL-based techniques for more accurate sentiment detection in e-commerce. Vadla et al. [44] present an innovative AI-driven approach to enhancing product design by leveraging SA of Amazon reviews using the BERT model. This research is pivotal in identifying customer emotions and preferences, which are essential for successful product innovation in the dynamic landscape of product design. The study introduces a prediction pipeline utilizing BERT and T5 models trained on synthetically generated and manually labeled datasets,

Table 1 Comparative analysis of selected studies

References	Dataset	Feature set	Method	Results
[10]	Yelp insurance reviews	AFINN, VADER	DT, KNN, SVM, LR, RF	Accuracy: 0.94
[14]	Food, mobile, and reviews	POS Tagging, SentiWordNet	Bayesian Mathematical Model	Accuracy: 0.87
[17]	Amazon reviews	Correlation, information gain	NB, DT, SVM, LR, NN, RF	Accuracy: 0.84
[20]	Conversational threads	–	KNN, DT, NB, ANN, LR, SVM	F1-score: 0.75
[24]	Hotel, airline, drug, book, automotive, office, music instrument, instant video	–	XLNet, RoBERTa, Bert, CNN, LSTM	F1-score: 0.61
[26]	Chinese E-Commerce product reviews	Contextual Embeddings	ERINE, BiLSTM	F1-score: 0.87
[28]	Drug product reviews	GLOVE, WORD2VEC	CNN, BERT, RoBERTa	Accuracy: 0.85
[34]	Amazon reviews	FastText, Word2Vec	CNN	Accuracy: 0.90
[36]	Online Yelp customer reviews	BoW	KNN, SVM, NB, BERT	Accuracy: 0.70
[40]	Tourist reviews	FastText	RoBERTaESP, GPT2, SVM	Accuracy: 0.99
[43]	Online product reviews	BoW, Word2Vec, TF-IDF, Contextual Embeddings	KNN, SVM, NB, BERT, XLNet	Accuracy: 0.89
[44]	Amazon reviews	Contextual Embeddings	BERT, T5	Accuracy: 0.91
[45]	IMDb	TextBlob, VADER, AFINN	BERT, LSTM, Sent WordNet, LR	F1-score: 0.92

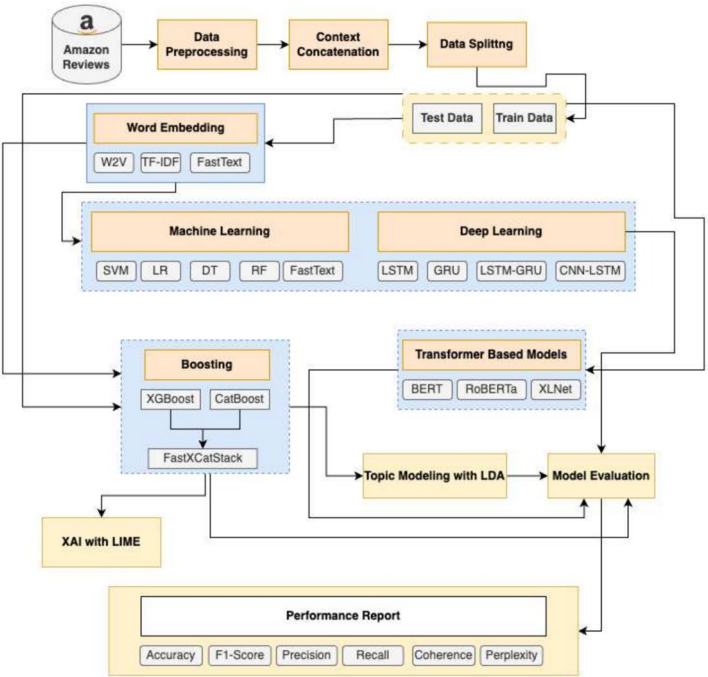

Fig. 2 Proposed work methodology

achieving remarkable accuracy rates-92% for BERT and 91% for T5. These models efficiently detect customer sentiments across different product aspects, thus providing invaluable insights for product design enhancements focused on eco-friendly products. The findings highlight the potential of AI in transforming product design by integrating customer feedback analysis into the development process, offering a substantial advantage in aligning product features with consumer expectations. Table 1 represents the comparative analysis of the current state-of-the-art methods.

After reviewing the existing literature, we conclude that many studies addressed SA using traditional ML and DL-based methods for online reviews. In this study, we will do this analysis using ML, DL, and transformers with different word embedding and hyperparameter tuning methods and in a different way that will provide us with a deep understanding of these approaches.

3 Methodology

In the initial stage, different models were used for predicting star ratings, while in the subsequent phase, the same models were utilized to address a tertiary classification problem. Following Fig. 2 represents our proposed work methodology,

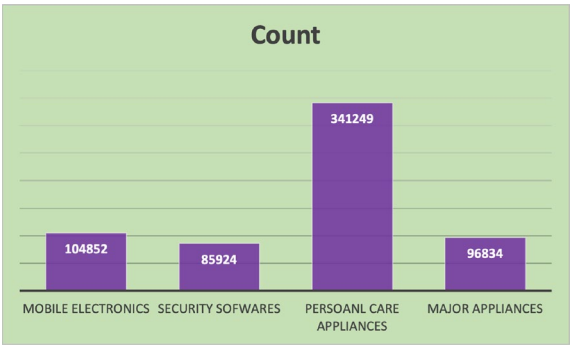

Fig. 3 Count of instances in datasets

3.1 Dataset

We experimented with our models on the Amazon product reviews dataset obtained from Kaggle,[4] focusing on star ratings ranging from 1 to 5. This range represents sentiments from the worst to excellent. Additionally, we utilized four publicly available datasets covering diverse domains, including mobile electronics, security software, personal care appliances, and major appliances. Figure 3 shows the number of instances in all four datasets.

Previous studies mostly focused on utilizing a singular input feature for their models. However, this approach often lacks the depth required for optimal performance and accurate predictions. Recognizing the limitations of relying solely on user reviews, our approach extends to gathering more comprehensive context and user sentiment related to specific products. This involves incorporating additional information from both the product title and headline. Hence, our study is grounded in product context-aware learning, aiming to enhance overall performance.

3.2 Data preprocessing

Data preprocessing serves to eliminate unnecessary text from the dataset, ensuring a streamlined and appropriate data format. In our study, we focus on four key columns: product title, review body, review headline, and star rating (which serves as the label column), discarding less relevant feature columns. We combine these three text columns and perform preprocessing which primarily involves converting uppercase letters to lowercase, eliminating irrelevant characters such as ASCII symbols, and tokenizing words, sentences, and stop-word removal. Additionally, we implement a filter using Python's RegEx library to identify and handle numbers, punctuation marks, and various patterns such as email addresses, URLs, and phone numbers. For transformer-based models, our preprocessing is confined to word and sentence tokenization, converting uppercase to lowercase, and eliminating irrelevant symbols. This limitation addresses the challenge of syntactic ambiguity [46], a

[4] https://www.kaggle.com/.

significant drawback in previous ML and DL-based algorithms. Syntactic ambiguity arises when words in a sentence can have multiple interpretations based on context, posing a challenging problem for interpretation.

3.2.1 Feature extraction

- **Product Titles and Descriptions** We extract key information from product titles and descriptions using natural language processing techniques. This includes tokenization, removal of stop words, and normalization of text to extract relevant keywords and phrases that represent the product context.
- **Review Headlines** Often summarizing the sentiment of the review, headlines are processed using similar NLP techniques to extract salient features that might influence the sentiment conveyed in the review body.

3.2.2 Incorporation into sentiment analysis model

- **Vectorization:** Extracted features are then vectorized using techniques such as TF-IDF, which helps in quantifying the weight of these context features in relation to SA. This method enhances the model's ability to recognize how specific product features relate to sentiment expressions in reviews.
- **Handling Mixed Sentiments with FastXCatStack** In addressing ambiguous or mixed sentiments in user reviews, our model leverages the nuanced understanding of textual context through advanced embedding techniques and contextual learning. For example, reviews that contain phrases like 'I loved the camera, but the battery life is too short' are challenging as they express mixed sentiments about different aspects of a product. Our approach, using FastXCatStack, which integrates FastText with XGBoost and CatBoost, effectively captures and weighs these mixed sentiments by analyzing the sentiment context associated with specific aspects of the products. This method enhances the model's accuracy in sentiment classification, particularly in distinguishing between overall positive, neutral, and negative sentiments even within single reviews.
- **Model Integration** These vectorized features are integrated into the ML-based models alongside traditional review text features. This integration allows the models to use contextual cues from the product features to better interpret the sentiment of the review.

3.2.3 Robustness of the feature extraction process

- **Validation** We validate the effectiveness of our feature extraction process through a series of tests that assess the impact of these features on model performance. This includes comparing model accuracy with and without the incorporation of product context features.
- **Reproducibility** The process is designed to be reproducible and consistent, with clear documentation and the use of standard NLP libraries and frameworks to ensure that other researchers can replicate our results.

3.3 Word embedding

Word embeddings offer numerical representations for textual inputs. Word embeddings transform words into vectors within a continuous vector space, allowing machines to understand semantic relationships and contextual nuances. This capability supports various natural language processing tasks, such as SA, text classification, and language modeling. By representing words as vectors, these embeddings help machines identify word similarities, grasp the meanings of words, and extrapolate from the data they've been trained on, thereby improving the effectiveness of different language-related tasks. In our research, we have implemented three distinct word embeddings-TF-IDF, Word2Vec, and FastText. Word2Vec embeddings consist of two models: Skip Gram (SG) and Continuous Bag of Words (CBoW). The CBoW outperformed SG when combined with different learning classifiers in our paper. TF-IDF demonstrated better performance compared to the Word2Vec model in our study, showing its effectiveness in capturing meaningful patterns and features from textual data.

FastText consistently demonstrates the best results in the proposed work, both when utilized independently as a supervised algorithm and when integrated with other DL and ML-based models. This robust performance highlights the effectiveness of FastText in enhancing the overall predictive capabilities of our study. FastText,[5] a word representation library developed by the Facebook research team, encompasses a repository of 2 million common crawl words with 300 dimensions, resulting in a vast collection of 600 billion word vectors. Distinguished by its unique approach, FastText incorporates hand-crafted n-grams as features alongside individual words [47]. FastText embedding utilizes morphological features, enhancing its ability to represent vectors and improving generalizability in various applications [48]. In this paper, FastText embeddings were obtained by training the model for 50 epochs with a learning rate of 0.005. After training the model, the quantization process is carried out using the *quantize* method, which relies on the *qnorm* and *retrain* parameters. This phase is vital for diminishing the model's size and possibly enhancing the speed of inference while maintaining accuracy with minimal impact. In ML, particularly with models such as FastText, quantization involves reducing the precision of a model's parameters, like weights and vectors. There are two main types: Post-Training Quantization and Quantization Aware Training. Our study focused on implementing Post-Training Quantization, which is applied after the model has been completely trained. This method does not necessitate the retraining of the model. However, some methods might employ a fine-tuning stage to compensate for any possible decreases in accuracy. Figure 4 represents the FastText word embedding architecture.

[5] https://fasttext.cc/.

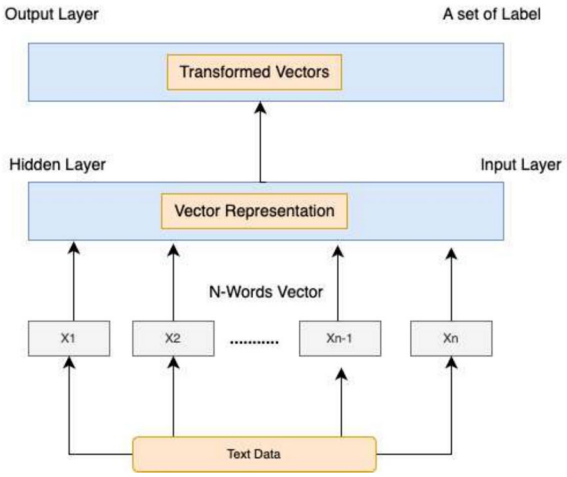

Fig. 4 FastText word embedding architecture

Table 2 Configuration details for ML models

Model	Regularization	Hyperparameter
DT	Split_{min}: [2, 5, 10]	GridSearchCV
RF	N-Estimators: [50, 100, 200]	GridSearchCV
SVM_{linear}	C: [0.1, 1, 10]	GridSearchCV
LR	C: [1, 10, 100]	GridSearchCV

3.4 Modeling approaches

In our study we implemented ML, DL, and transformer-based models, Following, the modeling approaches of this research work are summarized,

- **Support Vector Machine (SVM)** Chosen for its effectiveness in classification with high-dimensional data, often used in text classification due to its robustness and effectiveness in handling non-linear decision boundaries.
- **Logistic Regression (LR)** Utilized for its simplicity and efficiency in binary and multiclass classification problems, providing probabilistic outcomes that are beneficial for interpreting results in SA.
- **Decision Trees (DT) and Random Forests (RF)** Employed for their ability to handle large data sets with high dimensionality, providing insights into feature importance.
- **XGBoost and CatBoost** These boosting algorithms are known for their high performance in classification tasks and are used to enhance the predictive performance by combining multiple weak models into a strong classifier.
- **Deep Learning Models (LSTM, GRU, and CNN)** Selected for their ability to capture sequential patterns and dependencies in text data, essential for understanding the context in user reviews.

Table 3 Configuration details for boosting algorithms

Model	Rounds	lr	Depth	Sample	Function
XGBoost	100	0.1	3	0.8	logloss
CatBoost	100	0.1	6	0.7	logloss

- **Transformer-Based Models (BERT, RoBERTa, XLNet)** Incorporated due to their state-of-the-art performance in various NLP tasks, leveraging their deep bidirectional nature which is crucial for comprehending the nuances in textual SA.

3.4.1 ML based model

FastText embeddings were used as input for the subsequent supervised ML-based models, including DT, SVM, LR, and RF. These models were selected due to their varied capabilities in managing multiclass classification challenges. In addition, boosting methods such as XGBoost and CatBoost, along with feature engineering techniques, were applied. The following Table 2 represents the hyperparameters and regularization details for ML-based models in the proposed approach, for DT, the *Split_min* values (2, 5, 10) set the minimum samples needed to split a node, affecting complexity and the risk of overfitting. In RF, the $N - Estimators$ parameter (50, 100, 200) decides the forest's size, optimizing between speed and accuracy. SVM with a linear kernel and LR use the C regularization parameter (0.1, 1, 10 for SVM; 1, 10, 100 for LR) to adjust regularization strength, which is crucial for avoiding overfitting by limiting coefficient sizes. Higher C values mean less regularization and more complex decision boundaries.

These parameters were finely tuned through GridSearchCV, which conducts a comprehensive search over specified values to identify the best performance settings, ensuring each model achieves maximum accuracy and generalization.

We stacked FastText embeddings with XGBoost and CatBoost, leading us to name it FastXCatStack, which showed the best performance across all four datasets. The following Eqs. (1) and (2) represent the individual mathematical formulations for XGBoost.

$$\hat{y}_i^{(t)} = \phi(x_i) + \sum_{k=1}^{t} f_k(x_i), \quad f_k \in \mathcal{F} \tag{1}$$

where $\hat{y}_i^{(t)}$ is the prediction for the i-th instance at time t, $\phi(x_i)$ is the initial prediction for instance i, f_k represents an individual decision tree at iteration k, and $\mathcal{F}$ denotes the space of possible trees.

Next, we consider the CatBoost formulation:

$$\hat{y}_i(\mathbf{x}) = \text{softmax}\left(\phi_i(\mathbf{x}) + \sum_{k=1}^{K} f_{k,i}(\mathbf{x}) \right), \tag{2}$$

where $\hat{y}_i(\mathbf{x})$ is the predicted probability for class i based on the input vector $\mathbf{x}$, $\phi_i(\mathbf{x})$ represents the initial prediction for class i, and $f_{k,i}(\mathbf{x})$ is the output of the k-th decision tree for class i. Equation (3) stacks (1) and (2) for our proposed model FastXCatStack.

$$\hat{y}_i^S(\mathbf{x}) = \alpha \cdot \hat{y}_i^{(t)}(\mathbf{x}) + (1 - \alpha) \cdot \hat{y}_i(\mathbf{x}), \tag{3}$$

where $\hat{y}_i^S(\mathbf{x})$ is the final prediction for the i-th instance, combining predictions from XGBoost and CatBoost through a weighted sum where α is a weight parameter.

The following Table 3 represents the hyperparameters and regularization details for boosting methods in the proposed approach,

3.4.2 DL based model

In this study, we implemented two RNN-based models: LSTM with its variant BiLSTM and GRU with its variant BiGRU, and also their combined model LSTM-GRU. Additionally, we performed CNN with the stacking of LSTM using distinct layers. All these models achieved similar performance, just like ML-based models. Table 4 illustrates the hyperparameters and configuration details of each DL-based model. Notably, the count of each layer has been mentioned as well. This diverse array of models and configurations was chosen to thoroughly assess the potential of DL techniques in capturing complex patterns and relationships in the data, illustrating the flexibility and robustness of RNNs and CNNs in various configurations for challenging classification tasks.

3.4.3 Transformer based models

The Transformer is an NLP system designed to execute sequence-to-sequence processes following the self-attention mechanism with long-range dependencies comprising two main components encoder and decoder. Transformers were first introduced in 2017 by [49]. The self-attention mechanism, a core component of many modern neural network architectures, computes the representation of a sequence by relating different positions of the sequence to each other [50]. Mathematically, it is defined as:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK_i^\top}{\sqrt{d_k}}\right)V_i \tag{4}$$

In this equation, Q represents the query matrix that is used to compute the attention scores, K is the key matrix against which the queries are compared, and V is the value matrix from which the output of the attention mechanism is constructed. The term d_k denotes the dimensionality of the key vectors, which is used to scale the dot product thus stabilizing the softmax distribution. The index i refers to the position of a specific query vector within the sequence. The attention function outputs a weighted sum of the values, where the weight assigned to each value is determined by the compatibility of the query with the corresponding key, scaled by the square

Table 4 Configuration details for DL models

Model	Model layer	Dense layer	Dropout layer	Pooling layer	Flatten layer	Epochs	Function	Loss
LSTM-CNN	6	2	2	2	1	10	relu	Categorical entropy
LSTM	3	2	2	–	–	10	softmax	Categorical entropy
BiLSTM	4	2	2	–	–	10	softmax	Categorical entropy
GRU	3	2	2	–	–	10	softmax	Categorical entropy
BiGRU	4	2	2	–	–	10	softmax	Categorical entropy
LSTM-GRU	5	2	2	–	–	10	relu	Categorical entropy

Table 5 Configuration details for transformer based models

Model	Tokenizer	Batches	lr	Epoch
BERT	BertTokenizer	32	2e−5	5
XLNet	AutoTokenizer	32	2e−3	5
RoBERTa	AutoTokenizer	32	2e−5	5

root of the dimension of the key vectors, ensuring that the dot products do not grow too large in magnitude. This framework allows the model to dynamically emphasize different parts of the input sequence, making it particularly effective for tasks where relationships within the sequence are crucial, such as NLP and time series analysis.

This paper is based on utilizing transformers and focuses on optimizing their hyperparameters. Previous language models like RNNs, which were constrained by their computational and memory requirements for generative tasks, these transformers offer a significant advancement. In our study, we utilized large text datasets, and for this purpose, we employed text classification transformers such as BERT, XLNet, and RoBERTa. BERT, developed by Google, stands out for its deep bidirectional training, allowing the model to capture context from both directions (left and right) of a token within the text. This approach enables BERT to understand the nuances and complexity of language better than its predecessors, making it highly effective for a wide range of NLP-based tasks, including SA and question-answering.

Following BERT, XLNet introduces a permutation-based training strategy that allows it to learn the bidirectional context better by predicting the order of words within a sentence. This method helps XLNet to surpass BERT's limitations, particularly in contexts where the understanding of word order and relationships is crucial for comprehending the meaning of the text. RoBERTa, an iteration on BERT, modifies key hyperparameters in BERT and extends the training with more data for a longer period. These changes allow RoBERTa to achieve even better performance across various benchmarks. By training on larger datasets and optimizing the model's architecture, RoBERTa has demonstrated significant improvements in tasks requiring deep language understanding.

Table 5 represents the hyperparameters and configuration details for transformer-based methods in the proposed approach. Each model employs a specific tokenizer, with BERT utilizing the BertTokenizer and both XLNet and RoBERTa using the AutoTokenizer. The choice of tokenizer is critical as it prepares the input text data in a format that is compatible with each model, ensuring optimal processing and performance. For batch size, all models were trained using batches of 32. This size was selected to balance between computational efficiency and the ability to capture sufficient gradient information during training. The learning rate (lr) for BERT and RoBERTa was set to 2e−5, while XLNet utilized a higher learning rate of 2e−3. The difference in learning rates reflects the models' unique architectural nuances and training requirements, aiming to optimize each model's learning process for better convergence and performance.

Each model was trained for 5 epochs, which was determined to be the best duration for achieving peak performance based on our experiments. However, to thoroughly assess the models' robustness and to find the optimal training length, we also experimented with training durations of 3, 7, and 10 epochs. Despite these variations, the best results were consistently obtained at 5 epochs, indicating that extending training beyond this point did not yield significant improvements and, in some cases, could lead to overfitting. This detailed configuration and experimentation highlight our approach to fine-tuning the transformer-based models for optimal performance. By meticulously adjusting these parameters and training durations, we were able to maximize the efficiency and effectiveness of each model, with 5 epochs emerging as the ideal training length for achieving the highest accuracy in our text classification tasks.

4 Results and discussion

In our evaluation, standard metrics of accuracy, precision, recall, and f1-score were employed to quantify the model's classification performance.

4.1 Computational efficiency

To thoroughly assess the performance and efficiency of our proposed models, this study has conducted a detailed comparison with current state-of-the-art methods. The evaluation goes beyond traditional metrics such as accuracy, precision, recall, and F1-scores to also consider computational efficiency, an essential factor for practical implementations.

- **Hardware and Optimization** Our experiments were conducted on a MacBook M3 Max, equipped with 128GB of unified memory. This advanced setup allowed us to benchmark the computational requirements accurately, ensuring the reliability of our performance metrics across different modeling approaches.
- **ML Models Execution Time** On average, each epoch for our ML-based models requires approximately 5 min of execution time. This demonstrates the models' efficiency and suitability for scalable applications, highlighting their practical applicability in real-world settings.
- **DL-Based Models** DL-based models exhibited consistent performance, requiring about 5 min per epoch. This balance between computational demand and model efficacy underscores their potential in processing complex datasets efficiently.
- **Transformer-Based Models** Owing to their architectural complexity, transformer-based models necessitated about 15 min per epoch for training. Despite the longer duration, the significant enhancements in detection capabilities justify the computational investment, providing superior analytical depth and accuracy.

Table 6 Five classification results with ML for mobile electronics

Model	Precision	recall	Accuracy	f1-score
$SVM_{fasttext}$	0.83	0.84	0.84	0.84
$LR_{fasttext}$	0.83	0.83	0.83	0.83
$CATBoost_{fasttext}$	0.83	0.83	0.83	0.83
$XGBoost_{fasttext}$	0.82	0.83	0.83	0.82
$RF_{fasttext}$	0.82	0.82	0.82	0.82
$DT_{fasttext}$	0.69	0.69	0.69	0.69
$FastText_{supervised}$	0.68	0.71	0.71	0.68
$LR_{tf\text{-}idf}$	0.60	0.65	0.65	0.61
$SVM_{tf\text{-}idf}$	0.60	0.66	0.66	0.60
$RF_{tf\text{-}idf}$	0.60	0.66	0.66	0.60
$CATBoost_{tf\text{-}idf}$	0.60	0.66	0.66	0.60
$XGBoost_{tf\text{-}idf}$	0.60	0.65	0.65	0.60
$DT_{tf\text{-}idf}$	0.56	0.61	0.61	0.50
LR_{cbow}	0.62	0.67	0.67	0.63
SVM_{cbow}	0.61	0.66	0.66	0.60
$CATBoost_{cbow}$	0.60	0.66	0.66	0.60
$XGBoost_{cbow}$	0.60	0.65	0.65	0.60
RF_{cbow}	0.52	0.59	0.59	0.59
DT_{cbow}	0.43	0.42	0.42	0.43
LR_{sg}	0.61	0.66	0.66	0.61
SVM_{sg}	0.61	0.66	0.66	0.60
$CATBoost_{sg}$	0.59	0.59	0.60	0.60
$XGBoost_{sg}$	0.59	0.60	0.60	0.60
RF_{sg}	0.52	0.58	0.58	0.48
DT_{sg}	0.43	0.42	0.43	0.43

The evaluation scores for ML and DL-based methods, employing all three distinct word embedding techniques, are displayed in Table 6, specifically applied to a mobile electronics dataset. In this context, product title, user review, and headline concatenation were utilized to enhance contextual understanding of algorithms, aiming for optimal results and robust performance.

Analyzing Table 6, the evaluation of ML-based models for mobile electronics classification revealed a notable distinction in performance across various models and feature representations. Models utilizing FastText embeddings, specifically SVM, LR, and CATBoost, demonstrated superior performance, achieving top metrics in precision, recall, accuracy, and F1-score around 0.83 to 0.84. This indicates a highly effective balance between the precision of predictions and their relevance, coupled with an overall accuracy that underscores the models' ability to generalize well across the dataset. In contrast, models trained with TF-IDF and Word2Vec embeddings (both CBOW and skip-gram) showed significantly lower performance, with precision, recall, accuracy, and F1-scores dropping to as low as 0.43 in some

Table 7 Tertiary classification results with ML for mobile electronics

Model	Precision	Recall	Accuracy	f1-score
FastXCatStack	0.92	0.92	0.93	0.92
$CATBoost_{fasttext}$	0.91	0.92	0.93	0.93
$XGBoost_{fasttext}$	0.92	0.91	0.92	0.91
$SVM_{fasttext}$	0.92	0.91	0.92	0.91
$LR_{fasttext}$	0.92	0.91	0.92	0.91
$RF_{fasttext}$	0.92	0.91	0.92	0.91
$DT_{fasttext}$	0.86	0.86	0.86	0.86
$LR_{tf\text{-}idf}$	0.81	0.84	0.84	0.81
$SVM_{tf\text{-}idf}$	0.81	0.83	0.83	0.81
$CATBoost_{tf\text{-}idf}$	0.81	0.83	0.82	0.81
$XGBoost_{tf\text{-}idf}$	0.83	0.82	0.82	0.82
$RF_{tf\text{-}idf}$	0.80	0.80	0.80	0.75
$DT_{tf\text{-}idf}$	0.73	0.73	0.73	0.72
LR_{cbow}	0.80	0.83	0.83	0.80
SVM_{cbow}	0.82	0.83	0.83	0.79
$CATBoost_{cbow}$	0.78	0.78	0.78	0.78
RF_{cbow}	0.79	0.73	0.73	0.78
$XGBoost_{cbow}$	0.78	0.78	0.73	0.78
DT_{cbow}	0.66	0.66	0.66	0.66
LR_{sg}	0.80	0.82	0.82	0.80
$CATBoost_{sg}$	0.80	0.80	0.80	0.81
SVM_{sg}	0.78	0.77	0.77	0.72
$XGBoost_{sg}$	0.78	0.77	0.77	0.72
RF_{sg}	0.78	0.77	0.77	0.72
DTsg	0.65	0.64	0.64	0.64

cases, such as DT models with CBOW and skip-gram embeddings. The superior results of FastText can be attributed to its capability to leverage subword information, enhancing model understanding of the dataset's linguistic nuances. This contrasts with the more simplistic TF-IDF approach, which fails to capture semantic relationships, and the basic Word2Vec method, which, while capturing some context, lacks the morphological richness of FastText. Among the models evaluated, SVM, LR, and CATBoost with FastText embeddings emerged as the most effective, suggesting that the combination of advanced embedding techniques with robust classification algorithms can significantly improve text classification tasks. This highlights the importance of choosing the right feature representation and ML-based model to optimize classification performance in specific domains like mobile electronics reviews.

Additionally, this research study also addressed tertiary classification problems where ratings 1 and 2 were combined as 1 (indicating poor or not recommended), 3 was assigned as 2 (representing a neutral sentiment), and ratings 4 and 5 were grouped as 3 (indicating good or recommended). Subsequently, we integrated both

Table 8 Tertiary classification results with DL and transformer-based models for mobile electronics

Model	Precision	Recall	Accuracy	f1-score
$CNN - LSTM_{fasttext}$	0.91	0.91	0.92	0.91
$LSTM_{fasttext}$	0.91	0.91	0.91	0.91
$BiLSTM_{fasttext}$	0.91	0.91	0.91	0.91
$GRU_{fasttext}$	0.91	0.91	0.91	0.91
$BiGRU_{fasttext}$	0.91	0.91	0.91	0.91
$LSTM - GRU_{fasttext}$	0.91	0.91	0.91	0.91
$RoBERTa_{large}$	0.90	0.90	0.90	0.90
$BERT_{large\text{-}uncased}$	0.89	0.89	0.89	0.89
$BERT_{base\text{-}uncased}$	0.89	0.89	0.89	0.89
$XLNet_{large}$	0.86	0.87	0.87	0.86

three embeddings with the same ML-based models, as detailed in Table 7. The introduction of the FastXCatStack model, a combination of FastText embeddings with an ensemble of XGBoost and CATBoost models, achieved the highest performance metrics, showcasing precision, recall, accuracy, and F1-score at or above 0.92, with an accuracy peaking at 0.93. This model outperformed individual models trained solely on FastText embeddings, indicating the effectiveness of ensemble strategies in capturing nuanced sentiment distinctions. Models employing FastText embeddings consistently showed strong performance across all metrics, with precision and recall rates above 0.91 for most models, signifying their capability to accurately identify and classify sentiments as intended.

Tables 7 and 8 indicate that FastText embeddings surpassed all state-of-the-art models, even in tertiary classification problems. The performance metrics of DL models, specifically those utilizing combinations of CNN and RNN such as LSTM, BiLSTM, GRU, BiGRU, and hybrid LSTM-GRU architectures, uniformly achieved high precision, recall, accuracy, and F1-scores, all around 0.91. This consistency in performance across various DL architectures suggests that these models, when combined with FastText embeddings, are highly capable of capturing the sequential and contextual nuances of text data inherent in customer reviews, thereby effectively classifying sentiments into the designated categories. In contrast, transformer-based models like RoBERTa large, BERT large-uncased, BERT base-uncased, and XLNet large showed a slight decrement in their performance metrics, with precision, recall, accuracy, and F1-scores ranging from 0.86 to 0.90. Despite the transformers' advanced capabilities in understanding context and semantics through self-attention mechanisms, their performance in this specific task was marginally lower than that of the DL models paired with FastText embeddings. This could be attributed to the particular characteristics of the mobile electronics reviews dataset, where the nuanced understanding of domain-specific language and sentiment might be more effectively captured by models trained directly on the dataset with FastText embeddings. Comparatively, the ML ensemble model FastXCatStack and the CATBoost model with FastText embeddings previously mentioned achieved higher performance metrics, with precision, recall, accuracy, and F1-scores reaching up to

Table 9 Results from testing on the software review dataset

Model	Precision	Recall	Accuracy	f1-score
$SVM_{fasttext}$	0.90	0.91	0.91	0.90
$RoBERTa_{large}$	0.90	0.90	0.90	0.90
FastXCatStack	0.90	0.90	0.90	0.90
$CATBoost_{fasttext}$	0.90	0.90	0.90	0.90
$XGBoost_{fasttext}$	0.90	0.90	0.90	0.90
$LR_{fasttext}$	0.90	0.90	0.90	0.90
$RF_{fasttext}$	0.90	0.90	0.90	0.90
$CNN - LSTM_{fasttext}$	0.90	0.90	0.90	0.90
$BiGRU_{fasttext}$	0.90	0.90	0.90	0.90
$GRU_{fasttext}$	0.89	0.90	0.90	0.89
$BERT_{large\text{-}uncased}$	0.90	0.89	0.89	0.90
$LSTM_{fasttext}$	0.89	0.89	0.89	0.89
$BiLSTM_{fasttext}$	0.89	0.89	0.89	0.89
$LSTM - GRU_{fasttext}$	0.89	0.90	0.90	0.89
$BERT_{base\text{-}uncased}$	0.89	0.89	0.89	0.89
$XLNet_{large}$	0.87	0.87	0.87	0.87
$DT_{fasttext}$	0.84	0.83	0.83	0.83

Table 10 Results from testing on the major appliances dataset

Model	Precision	Recall	Accuracy	f1-score
FastXCatStack	0.93	0.93	0.93	0.93
$SVM_{fasttext}$	0.93	0.93	0.93	0.93
$CATBoost_{fasttext}$	0.93	0.93	0.93	0.93
$XGBoost_{fasttext}$	0.93	0.93	0.93	0.93
$CNN - LSTM_{fasttext}$	0.93	0.93	0.93	0.93
$LSTM - GRU_{fasttext}$	0.93	0.93	0.93	0.93
$BiLSTM_{fasttext}$	0.93	0.93	0.93	0.93
$LSTM_{fasttext}$	0.93	0.93	0.93	0.93
$RF_{fasttext}$	0.93	0.93	0.93	0.93
$BiGRU_{fasttext}$	0.92	0.93	0.93	0.92
$LR_{fasttext}$	0.92	0.93	0.93	0.92
$GRU_{fasttext}$	0.92	0.92	0.92	0.92
$DT_{fasttext}$	0.89	0.88	0.88	0.89
$RoBERTa_{large}$	0.92	0.92	0.92	0.92
$BERT_{large\text{-}uncased}$	0.92	0.91	0.92	0.92
$BERT_{base\text{-}uncased}$	0.90	0.90	0.89	0.90
$XLNet_{large}$	0.89	0.89	0.88	0.90

0.92 and 0.93, respectively. This indicates that for the specific task of tertiary classification in mobile electronics reviews, ensemble methods and carefully tuned ML-based models with advanced feature representations may offer a slight edge over DL and transformer-based models.

Table 11 Results from testing on the personal care appliances

Model	Precision	Recall	Accuracy	f1-score
FastXCatStack	0.94	0.94	0.94	0.94
$CATBoost_{fasttext}$	0.94	0.93	0.93	0.94
$XGBoost_{fasttext}$	0.93	0.94	0.94	0.93
$SVM_{fasttext}$	0.93	0.93	0.93	0.93
$CNN - LSTM_{fasttext}$	0.93	0.93	0.93	0.93
$LSTM - GRU_{fasttext}$	0.93	0.93	0.93	0.93
$BiGRU_{fasttext}$	0.93	0.93	0.93	0.93
$GRU_{fasttext}$	0.93	0.93	0.93	0.93
$LSTM_{fasttext}$	0.93	0.93	0.93	0.93
$RF_{fasttext}$	0.93	0.93	0.93	0.93
$BiLSTM_{fasttext}$	0.92	0.92	0.92	0.92
$LR_{fasttext}$	0.92	0.92	0.92	0.92
$RoBERTa_{large}$	0.92	0.92	0.92	0.93
$BERT_{large\text{-}uncased}$	0.93	0.91	0.92	0.92
$BERT_{base\text{-}uncased}$	0.90	0.90	0.89	0.90
$XLNet_{large}$	0.90	0.90	0.90	0.90
$DT_{fasttext}$	0.88	0.88	0.88	0.88

The results suggest that while DL and transformer-based models are powerful tools for text analysis, their effectiveness can vary depending on the nature of the task and the characteristics of the data. Moreover, the slight edge of the FastXCat-Stack and CATBoost models underscores the importance of feature representation and model architecture selection in achieving optimal performance in sentiment classification tasks. These insights point towards the nuanced trade-offs between different approaches and the potential for hybrid or ensemble models to leverage the strengths of multiple methodologies for improved performance in complex classification scenarios.

In this section, the rest of the work has been performed on testing for the other three datasets which are software reviews, major appliances, and personal care appliances presented in Tables 9, 10 and 11.

The evaluation of the Software Review Dataset in Table 9, the results showcased a high degree of proficiency across a diverse array of models, including both traditional ML with FastText embeddings and advanced DL and transformer-based models. Notably, almost all models achieved precision, recall, accuracy, and F1-scores in the vicinity of 0.89 to 0.90, indicating a successful adaptation to the dataset's intricacies, which are characterized by technical terminology and nuanced expressions typical of software reviews. Among RNN-based models, LSTM, BiLSTM, and their variants, including hybrid models like LSTM-GRU, slightly trailed with scores around 0.89, which still signifies a commendable level of performance. Their slightly lower scores compared to some other models could indicate the challenges of capturing the full extent of semantic relationships in software reviews using solely sequence-based processing. The DT model, even when enhanced with FastText embeddings, exhibited the most significant drop in

performance, with scores around 0.83. This could highlight the model's limitations in dealing with the complexity and high dimensionality of text data, underscoring the necessity for more complex model structures to navigate the intricate patterns found in software reviews effectively. The proposed FastXCatStack model again achieved the F1-score of 0.90 while the SVM with FastText exhibited the best performance in the software review dataset. This analysis reveals a crucial insight: the nuanced understanding of text, particularly in specialized domains like software reviews, benefits significantly from sophisticated modeling techniques and advanced text representations. The close performance across various advanced models suggests a convergence towards effective methodologies for text classification but also indicates room for designed approaches, especially when addressing the unique challenges presented by specific types of textual data.

The evaluation of various models in Table 10 on the Major Appliances Dataset showcases an impressive uniformity in performance leveraging FastText embeddings, which have achieved remarkable precision, recall, accuracy, and F1-scores, predominantly clustering at 0.93. This consistency in high performance across a spectrum of models, including SVM, CATBoost, XGBoost, and various DL-based configurations such as CNN-LSTM, LSTM-GRU, and BiLSTM, as well as traditional ML-based models like RF, indicates a strong alignment between the models' capabilities and the dataset's characteristics, which likely includes distinct patterns and terminologies specific to major appliances reviews. Notably, the ensemble model FastXCatStack and the SVM with FastText embeddings stand out, affirming the strength of combining robust text representations with powerful ML-based algorithms. These models excel in understanding and classifying the nuanced language typical of major appliance reviews, highlighting the importance of feature representation in achieving high accuracy. The slightly lower performance of BiGRU, LR, and GRU models, with scores around 0.92, still reflects a high level of competency, though it suggests a minor divergence in model efficiency in capturing all nuances of the dataset. DT, despite utilizing FastText embeddings, lags slightly behind, with scores around 0.88 to 0.89, underscoring the model's limitations in handling complex, high-dimensional text data. Transformer-based models like RoBERTa, BERT (both large-uncased and base-uncased), and XLNet demonstrated varying degrees of success, with precision, recall, accuracy, and F1-scores ranging from 0.88 to 0.92. These models, known for their deep contextual understanding, offer insights into the dataset's complexity but appear slightly less effective than the leading models. The marginally lower performance of these transformers, compared to FastText-based models, might be attributed to the specific challenges and characteristics of the Major Appliances Dataset, possibly indicating that the context captured by transformers does not translate into a significant advantage over the semantic and syntactic richness encapsulated by FastText embeddings in this particular domain. The results from the Major Appliances Dataset affirm the critical role of advanced feature representations, like FastText, in enhancing model performance. Moreover, they highlight the potential of combining these representations with both traditional and advanced ML techniques to optimize classification outcomes, particularly in specialized domains requiring a nuanced understanding of the text.

For the Personal Care Appliances dataset in Table 11, the evaluation results indicate a strong performance across a wide array of models, particularly highlighting the effectiveness of FastText embeddings in enhancing text classification tasks. The FastXCatStack model, alongside individual CATBoost, XGBoost, and SVM models utilizing FastText, achieved remarkable precision, recall, accuracy, and F1-scores, all peaking at 0.93 or 0.94. This suggests a high level of alignment between the models' capabilities and the dataset's unique characteristics, likely marked by specialized terminology and consumer sentiments specific to personal care appliances. The uniform success of models across the board, including DL configurations like CNN-LSTM, LSTM-GRU, BiGRU, and more traditional algorithms such as Random Forest, equipped with FastText embeddings, underscores the pivotal role of sophisticated text representations. These embeddings, by capturing the nuanced semantics and syntactical patterns of the dataset, significantly contribute to the models' ability to discern and classify varying sentiments accurately. In comparison, transformer-based models such as RoBERTa and BERT (both large-uncased and base-uncased), as well as XLNet, showed a slight variance in their performance, with scores ranging from 0.89 to 0.93. While still competitive, the marginally lower metrics compared to the leading models could suggest that the advanced contextual understanding provided by transformers does not translate into a substantial advantage for this specific dataset over the semantic richness captured by FastText embeddings.

The DT model, even with FastText, displayed the most noticeable drop in performance metrics, hovering around 0.88. This outcome likely reflects the limitations of simpler models in processing the complex and high-dimensional data characteristic of text classification tasks, underscoring the necessity for more elaborate model structures to handle the intricate patterns present in personal care appliance reviews effectively. The analysis of the Personal Care Appliances dataset results reiterates the critical importance of selecting appropriate feature representations and modeling techniques to achieve optimal classification outcomes. The high performance across most models suggests that current methodologies are well-suited to tackle the challenges posed by specialized domains, with advanced embeddings and ensemble strategies offering a slight edge in navigating the nuanced landscape of consumer reviews.

From the testing results, it is evident that FastXCatStack achieved the highest accuracy scores on three datasets: mobile electronics (0.93), major appliances (0.93), and personal care appliances (0.94). Meanwhile, SVM with FastText also demonstrated notable results, securing the top accuracy score of 0.91 on the Software Reviews Dataset.

Figure 5 illustrate the confusion matrix and classification matrix by class for tertiary classification SA in the personal care appliances dataset.

5 Comparison of the results with the state-of-the-art

In this section, we compare our results with three baseline studies, [43, 44], and a third study [34]. The study by Shaik et al. [44] utilizes BERT and T5 models to analyze sentiment from Amazon reviews, specifically focusing on eco-friendly

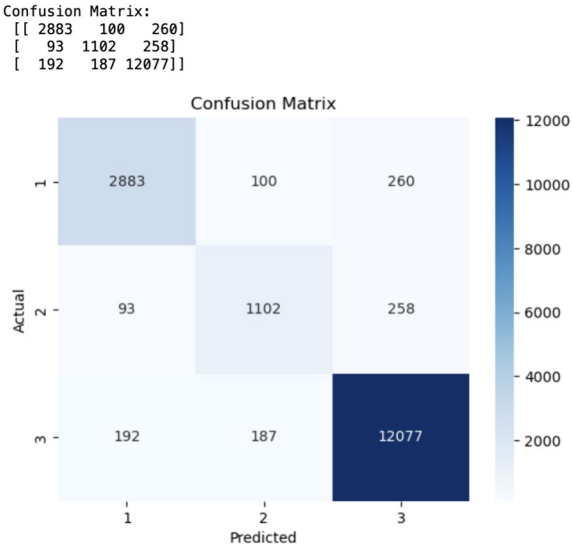

Fig. 5 Confusion matrix for FastXCatStack

Table 12 Comparison of the results with the state of the art

Refs.	Dataset	Baseline score	Proposed work
[34]	Amazon reviews	Accuracy: 0.90	Accuracy: 0.92
[43]	Amazon reviews	Accuracy: 0.89	Accuracy: 0.92
[44]	Amazon reviews	Accuracy: 0.91	Accuracy: 0.93

products. Their approach achieved an accuracy of 91% for T5, which was particularly noted for its depth of aspect-based analysis and its application towards enhancing product design. Ali et al. [43] explore the effectiveness of various ML, DL, and transformer-based algorithms in extracting actionable insights from customer reviews, achieving an overall accuracy of 89%. This study emphasizes the importance of integrating customer feedback into the design process to ensure products meet user expectations. The third study by Qorich et al. [34] investigates the use of CNNs and LSTM networks to understand consumer sentiments across several product categories on Amazon, achieving an accuracy of 90%. This study highlighted the potential for hybrid models to capture nuanced emotional expressions in text data. Our results, which show an improvement over these baseline studies, are presented in the Table 12. This table illustrates that our proposed model not only enhances accuracy but also ensures consistency across varied datasets, highlighting robustness against differing data characteristics. The enhancements in accuracy from 0.90 to 0.92, from 0.89 to 0.92, and from 0.91 to 0.93 in comparisons to the studies by [34, 43, 44] respectively, indicate our model's superior capability in handling complex patterns in data which are often missed by conventional models. These improvements underscore our method's potential for real-world applications where such gains can significantly impact performance and outcomes.

Table 13 LDA results across all datasets

Dataset	Coherence	Perplexity
Mobile electronics	0.47644	−7.8241
Software reviews	0.39076	−7.7566
Personal care appliances	0.49574	−7.9641
Major appliances	0.31678	−7.5455

6 Interpretability modeling

6.1 LDA

Interpretability modeling involves creating models or methods that enhance the understanding and transparency of complex ML models. LDA is widely acknowledged as a model that strikes a positive balance between interpretability and complexity in the domain of topic modeling. It has been designed to uncover topics in a collection of texts. These topics can be interpreted as sets of words that frequently co-occur [51]. We performed the LDA algorithm to extract the top four most similar features in all three datasets.

To evaluate the LDA results, we implemented coherence and perplexity as key metrics. Perplexity measures model prediction accuracy, while coherence measures topic interpretability. These metrics were instrumental in assessing the quality of our LDA model and guiding our analysis [52].

In this study, Table 13 shows that the LDA model achieved scores of 0.47644, 0.39076, 0.49574, and 0.31678, respectively. These scores indicate the model's robust understanding of the more detailed and subtle aspects of the data. Furthermore, the negative sign in perplexity values results from taking the logarithm in its formula, with lower negative values indicating improved language model performance on the data.

6.2 LIME

LIME, a model-agnostic approach, provides interpretable explanations for complex black-box models, enabling users to understand how these models make decisions on individual instances. This approach provides insights into how the predictions made by a model align with the specific demands of a particular task. It proves particularly valuable in contexts where understanding the rationale behind a model's decisions is equally important as the precision of its results. In Eq. 5, the objective of the LIME methodology is to find an easily understandable model, symbolized as $\hat{g}$, within a specified set of models G. The goal is to reduce the loss $\mathcal{L}$, which quantifies the difference between the predictions made by the model g and those of the more complex model f, considering the influence of the locality kernel π_x. Moreover, $\Omega(g)$ denotes the complexity of the model g, highlighting a preference for simpler models to enhance interpretability and maintain straightforwardness.

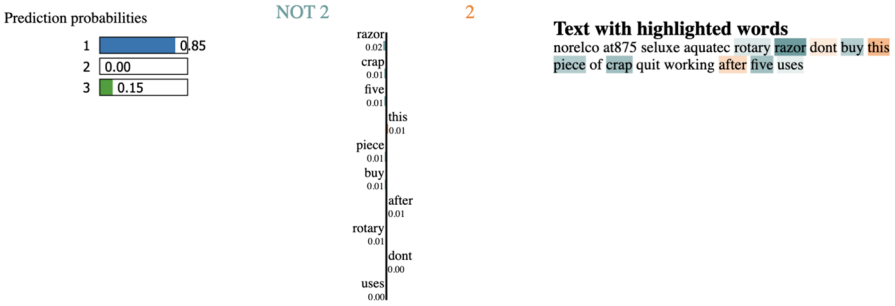

Fig. 6 (1–2) Star—not recommended

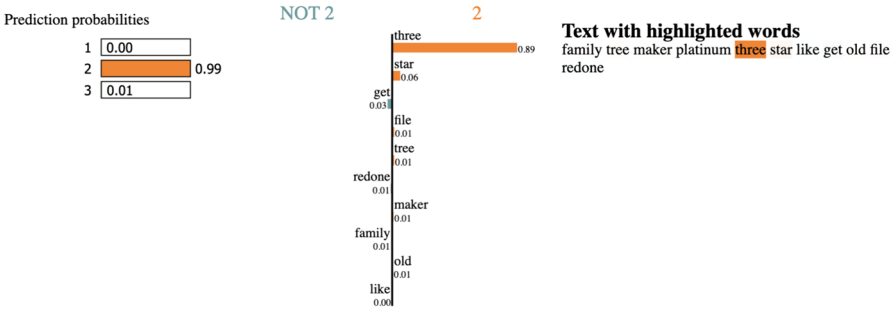

Fig. 7 (3) Star—neutral

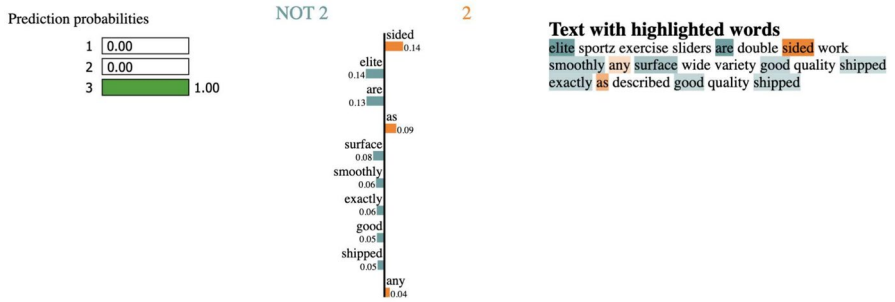

Fig. 8 (4–5) Star—recommended

$$\hat{g} = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{5}$$

Figure 6 illustrates a user's negative sentiment towards a "rotary razor", aligning with our classification for class 1, representing either a 1 or 2-star rating and indicating a not recommended product. LIME accurately identifies this sentiment by highlighting specific terms such as "crap", "piece", and "five" as evident in the figure. These terms are critical in understanding why the algorithm classified the sentiment as negative, demonstrating LIME's capability to reveal the specific

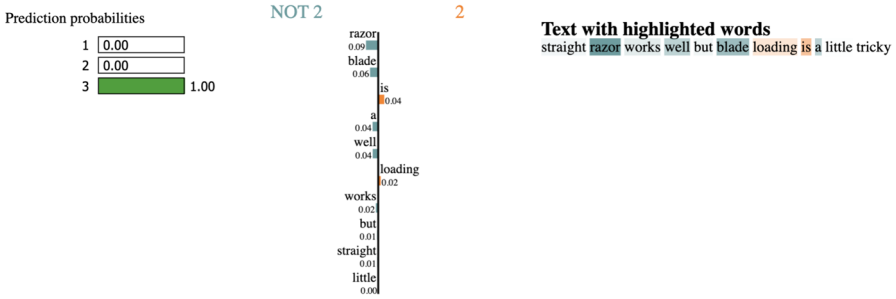

Fig. 9 Neutral but predicted as recommended

language that influences its classification decisions. Similarly, in Fig. 7, the sentence conveys a neutral sentiment, and LIME accurately identifies the neutral term "three," which does not express a positive or negative sentiment. The term "three" likely refers to a middle-of-the-road rating, such as a 3-star review, which does not convey strong feelings in either direction. This precise identification of neutrality by LIME underscores its utility in dissecting and interpreting the sentiment conveyed in a text, offering clear insight into the more subtle aspects of language that inform its analytical processes. Moving on to the Fig. 8, where the sentiment is strongly recommended, LIME provides insightful explanations by emphasizing positive words like "elite", "exactly", "good", and, "smoothly". These terms, associated with high quality and satisfaction, directly contribute to the model categorizing the review under a class that likely corresponds to a high rating, such as 4 or 5 stars, denoting strong recommendations.

However, in the case of Fig. 9, a wrong prediction occurred; the actual class label was neutral, but LIME misidentified it as recommended, pointing to terms like "razor", "blade", "well", and others. This instance illustrates a limitation or challenge in the interpretative process, where the context or the subtle nuances of how certain terms are used within the review might not have been fully captured or weighed accurately by the model. Such inaccuracies underscore the complexity of SA, especially in instances where specific keywords might generally be associated with positive or negative sentiments but, in context, do not necessarily convey a strong sentiment or the sentiment intended by the user. These examples underscore the effectiveness of our proposed FastXCatStack in delivering superior performance. We implemented LIME to interpret this performance, gaining deep insights into model predictions and ensuring a nuanced understanding of the sentiments expressed in user reviews.

7 Conclusion and future work

This research introduces a robust hybrid approach for SA, integrating product context learning across four distinct Amazon review datasets. The proposed FastXCatStack model provided us with the highest accuracy and F1-score, outperforming

Table 14 Topics and items for four datasets

Dataset	Topic	Items
Mobile electronics	Electronics Accessories	Cable, battery, bluetooth, case, cover
	Audio devices	Speaker, sound, radio, ipod
	Apple products	Apple, iphone, ipod, ipad, touch
Software reviews	Software categories	Microsoft, windows, mac, upgrade, version
	Office and productivity software	Product, work, upgrade, office, home
	Antivirus and security software	Norton, internet, computer, tax, program
Personal care appliances	Hair care	Hair, shaver, razor, love, best
	Body care	Massage, body
	Muscle care	Muscle, size, energy
Major appliances	Kitchen appliances	Hood, fridge, dishwasher, refrigerator
	Appliances maintenance and services	Service, year, unit, time, door
	Appliance performance	Great, range, filter, fit, star

alternative embedding methods, including DL-based models and state-of-the-art transformers. Notably, the FastText embedding proved valuable, particularly in the context of sentence-level SA within large datasets. Providing the learning algorithm with additional context has significantly improved performance and robustness across diverse domains.

Despite these insights, this study is confined to Amazon product reviews, limiting generalizability to other types of reviews or content from different platforms. The specificity of product reviews and the nature of Amazon's platform may influence the applicability of our findings to other e-commerce or review contexts. Additionally, our analysis is limited to texts written in English, which restricts the applicability of our models to global datasets that include multilingual content. The product context significantly influences SA results by providing additional descriptive details that enhance the understanding of user sentiments. For instance, understanding whether a product belongs to 'high-end electronics' or 'budget appliances' can shift the sentiment interpretation of words like 'expensive' or 'cheap.' However, our model's performance can vary across different product categories due to the variability in how consumers express sentiments in various contexts.

Moving forward, we are motivated to expand our research to handle various informal texts in different languages across diverse domains. We are determined to utilize advanced transformers such as T5, GPT models, and others, aiming to solve problems related to multi-label and multi-classification issues across diverse languages, including those with limited resources. Future work will also be oriented towards generalizing and enhancing the applicability of AI-driven models for SA. Specifically, the research will extend beyond Amazon product reviews to encompass a broader range of review platforms and types of content. This will address the current confinement to a single platform and type of review, thereby increasing the generalizability of our findings.

Appendix

The following Table 14 presents the distribution of topics and associated items identified through LDA across four distinct consumer product datasets, including Mobile Electronics, Software Reviews, Personal Care Appliances, and Major Appliances.

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Author Contributions EH: conceptualization, data analysis, formal analysis, research execution, design of methods, resources, software, writing original draft, investigation. SYY: visualization, supervision, project management, funding acquisition, research conduct, validation.

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Data availability and access The data utilized for this study is publicly available and can be accessed on Kaggle(https://www.kaggle.com/datasets/cynthiarempel/amazon-us-customer-reviews-dataset?select=amazon_reviews_us_Major_Appliances_v1_00.tsv).

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical and informed consent for data used Not applicable.

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